$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \tag{8.37}$$

$$\dot{\boldsymbol{p}} = \boldsymbol{F}(\boldsymbol{q}) - \frac{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{F}(\boldsymbol{q})}{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}}\boldsymbol{p}. \tag{8.38}$$

These equations preserve $\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} =$ constant. (Note: It is tempting to replace the denominator in this expression by K_0, the value of kinetic energy, but this is not recommended since it then destroys the first integral.) The system (8.37)–(8.38) can be rewritten in the alternative form

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p},$$
$$\dot{\boldsymbol{p}} = [I - \mathcal{H}]\,\boldsymbol{F}(\boldsymbol{q}),$$

where $\mathcal{H} = \boldsymbol{p}(\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p})^{-1}\boldsymbol{p}^T\boldsymbol{M}^{-1}$ is a *projector* matrix which is idempotent: $\mathcal{H}^2 = \mathcal{H}$. The invariant subspace associated to this projector is span$(\boldsymbol{p})$:

$$\mathcal{H}\boldsymbol{p} = \boldsymbol{p}.$$

Isokinetic dynamics has nonvanishing divergence, which requires a special treatment to elucidate its statistical mechanical properties [338, 376]. In order to find a stationary solution, one substitutes

$$\rho(\boldsymbol{q},\boldsymbol{p}) = e^{-w(\boldsymbol{q},\boldsymbol{p})}\Phi(K(\boldsymbol{p})), \tag{8.39}$$

(with Φ a nonvanishing function of K) into the continuity equation for the isokinetic system, the function w being at this point unknown. Then one obtains, with $\boldsymbol{f}_{\text{iso}}$ denoting the vector field in (8.37)–(8.38), that

$$\begin{aligned} 0 = \nabla\cdot(\rho\boldsymbol{f}_{\text{iso}}) &= (\nabla\rho)\cdot\boldsymbol{f}_{\text{iso}} + \rho(\nabla\cdot\boldsymbol{f}_{\text{iso}}) \\ &= (-\nabla w\rho + \rho\frac{\Phi'(K)}{\Phi(K)}\nabla K)\cdot\boldsymbol{f}_{\text{iso}} + \rho(\nabla\cdot\boldsymbol{f}_{\text{iso}}). \end{aligned}$$

Using the isokinetic property, the term involving K drops out. One next calculates the divergence of the isokinetic vector field and finds that (see Exercise 5)

$$\nabla\cdot\boldsymbol{f}_{\text{iso}} = (\mathrm{N_c} - 1)\frac{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{F}}{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}}.$$